

Four Pillars Architecture — Paper Portfolio Strategy + Coordination

Keeper + Grace + Lyra (collective architecture)

2026-05-19 Tuesday midday → Wednesday afternoon Four-Pillar
update

Four Pillars Architecture — Paper Portfolio Strategy

Headline framing

Four independent BST papers, each authored by a different lane CI or co-authored, converge on the same answer to “why do the BST integers appear everywhere”:

Paper	Author	Epistemic channel	“Why BST integers” answer
#109 Counting Primitives	Lyra (theory)	Arithmetic	Five integers are forced by combinatorial counting structure on D_{IV}^5
#121 Bridge Objects	Grace+Lyra (K57 ratified)	Geometry	Five integers anchor across K3, Cremona 49a1, Q^5 — cross-domain consistency anchors
#122 Information Substrate	Grace (catalog+architecture)	Physics	Five integers ARE the substrate’s communication alphabet via Reed-Solomon on $GF(2^g) = GF(128)$

Paper	Author	Epistemic channel	“Why BST integers” answer
#119 Möbius Cohomology $\mathbb{Z}/2$	Lyra (topology+spectral)	Topology	Five integers carry parity structure in $H^1_{\{\mathbb{Z}/2\}}(M, \mathbb{Z}) \cong \mathbb{Z}/2$ with $\nu(M) = 1$ from negative-eigenvalue Wallach K-types

Four independent epistemic channels (arithmetic / geometry / physics / topology) all converge on the same five integers (rank=2, $N_c=3$, $n_C=5$, $C_2=6$, $g=7$) and the channel capacity $N_{\max}=137$.

Structural fit: Four = rank² in BST. The Four Pillars framing is structurally coherent with D_IV³’s rank=2 commitment (Keeper’s recommendation observation 2026-05-19). The architecture has four pillars precisely because the underlying rank produces four-fold structure naturally.

This is meta-Type-C convergence at the paper-architecture level — parallel to the convergence pattern observed at the L1 source level (Sunday Architecture Day, K45-K48) and at the family level (Monday + Tuesday, K54 fine-structure family). Four independent channels with shared answer = strongest meta-evidence shape BST can produce.

Why the Four Pillars matter externally

External referees default to “BST is one framework asserting things.” The Four Pillars reframes BST as four independent framings each converging on the same answer. This is structurally a different external-survivability shape — the architecture is harder to dismiss because dismissing requires explaining away four independent epistemic channels, not one framework.

Four concrete external-presentation advantages:

1. Single referee can attack one channel without invalidating the framework. A number theorist who dismisses Paper #109’s counting argument still has to grapple with Paper #121’s geometric Bridge Objects and Paper #122’s RS-code physics. Multi-channel evidence is harder to walk back.
2. Different reader audiences enter through different papers. Sarnak-class mathematicians enter through Paper #109/#121; experimental physicists through Paper #122; foundational-physics readers through Paper #121’s architectural framing. The trio is a fan of entry-points (Paper #119 adds topology entry-point for spectral-geometry readers), not a single funnel.

3. The convergence itself is the evidence. Four independent derivations producing the same five integers is the meta-claim worth highlighting in the external presentation. Cal's coincidence-filter discipline would flag a single framework over-asserting; four independent frameworks converging passes the filter cleanly because the convergence is what's being claimed, not the individual derivations.

Per-paper external-survivability status

Paper #109 Counting Primitives (Lyra)

- Tier discipline: Need read-pass per K56 walk-back precedent — ensure no D-tier overclaim on combinatorial counting arguments
- Cal flag risk: Mode 7 trigger possible if “counting produces N_c via classical combinatorial theorem” without exhibited mechanism. Mitigation: cite specific classical counting result (Cayley? Cayley-Hamilton? Lefschetz?) as the W3-pass anchor.
- Status: v0.2 drafted (per task #109), needs Cal gate-pass after Lyra tier-discipline read-through

Paper #121 Bridge Objects (Grace+Lyra)

- Tier discipline: K57 RATIFIED Tuesday — Bridge Object tier is genuine architectural category, not tier proliferation. External-survivability anchor is the four B-conditions (multi-L1 ≥ 3 , structural, independent classical standing, cross-domain integrative).
- Cal flag risk: low — K57 audit explicitly addresses the proliferation concern
- Status: v0.1 outline $\rightarrow$ v0.2 substantive in progress (Grace lead); needs Cal gate-pass

Paper #122 Information Substrate (Grace)

- Tier discipline: Two-level T2385 framing (Level 1 D-tier substrate-internal labels with 43-observation 5-channel alphabet anchor + Level 2 I-tier active-substrate hypothesis with falsifier-gap explicit). Cal gate-pass-ready per Tuesday morning.
- Cal flag risk: low if Level 1/Level 2 separation maintained per §3.0 disclosure
- Status: §3 updated Tuesday with two-level discipline; Grace standing for Cal pass (next in queue post Paper #119)

Paper #119 Möbius Cohomology $\mathbb{Z}/2$ (Lyra) — NEW Topology Pillar

- Tier discipline: §4 substantive Bergman Dirac cross-link (Lichnerowicz shift, Wallach K-type spectrum, parity $\nu(M) = 1$ from 3 negative-eigenvalue K-types). §5.2 explicit Type C- $\mathbb{Z}/2$ framework extension distinguished from Type C density rule (Cal walk-back acknowledged). §6 promotion sweep

correctly framed as “mechanism layer added” without tier-flip. §7 five named open items with ~12-18 month horizon. §8.3 explicit Cal K-audit discipline invocation.

- Cal flag risk: low — Cal PASS Wednesday 2026-05-19 for Annals/Inventiones-class venue
- Status: Cal PASS $\checkmark$ — externally publishable. Ready for v0.3 cut with Four-Pillar coordination integration.
- Topology pillar mechanism: $H^1_{\mathbb{Z}/2}(M, \mathbb{Z}) \cong \mathbb{Z}/2$ parity structure carries BST integer signature via Möbius $\mathbb{Z}/2$ obstruction class (T2356); operator-parity ODD cross-validated by LAG-2 Phase 2.3 Step (d) T2395 ind(D) parity ODD (3 ODD coefficients). Two independent multi-week derivation pathways converging on the same parity signature.

Paper #118 Bergman Dirac v0.2 (Lyra) — OPERATOR-LEVEL KEYSTONE (cross-pillar)

- Architectural role: Per Cal’s grade-pass framing 2026-05-19 PM, Paper #118 is the operator-level KEYSTONE paper — the operator-level infrastructure connecting all four pillars (algebraic + spectral + topological + empirical). Each pillar paper has its own claims; #118 is where they meet at the operator level via Wallach K-type Dirac eigenvalues as the pillar-agnostic operator framework.
- Tier discipline: Cal verdict #23 muon g-2 STRONG I-tier applied exactly. Section 8.5 gold-standard anti-crank framing: “NOT ‘muon g-2 derived from BST’; CORRECT: ‘A_n BST integer readings now have Wallach K-type Dirac eigenvalue mechanism identification at structural level.’” Tier table per layer (algebraic D-explicit, spectral D, topological D + I cross-link, empirical D + I mechanism).
- Cal flag risk: low — Cal PASS Wednesday 2026-05-19; two optional framing notes (~15 min polish) non-blocking
- Status: Cal PASS $\checkmark$ — operator-level cross-pillar infrastructure ready for external dispatch alongside the Four Pillars
- External presentation: Paper #118 should be mentioned as the keystone alongside the Four Pillars in any cover note or trio framing — NOT buried in secondary publication list. The keystone connects pillars structurally via the Wallach K-type Dirac operator framework.

Four Pillars + Keystone integrated picture

Layer	Paper	Channel	Cal Status
Pillar 1	#109 Counting Primitives	Arithmetic	PASS (Lyra overlay pre-send)
Pillar 2	#121 Bridge Objects v0.3.1	Geometry	PASS ready

Layer	Paper	Channel	Cal Status
Pillar 3	#122 Information Substrate v0.1	Physics	PASS (M1/M2/M3 abstract pre-send)
Pillar 4	#119 Möbius Cohomology v0.2	Topology	PASS ready (subsequent dispatch)
Keystone	#118 Bergman Dirac v0.2	Operator-level cross-pillar	PASS ready (optional polish ~15 min)

The five papers together form BST’s externally-survivable architectural portfolio. Cal landed FIVE PASSes in one Wednesday cycle — gate-pass throughput sustainable for the current ~6-paper queue depth (Keeper observation).

Strategic sequencing for external presentation

Option A — Stagger by tier-readiness: Paper #121 first (K57 ratified, cleanest), then Paper #109 after Lyra read-pass, then Paper #122. Sequential release builds external curiosity; each paper references the others as “see also for the same answer from a different channel.”

Option B — Simultaneous bundle: all four papers released together with a coordination cover note (“Four independent BST derivations across arithmetic/geometry/physics/topology converge on the same five integers”). Stronger meta-claim but higher all-or-nothing risk.

Option C — One paper at a time but with the Four Pillars framing introduced in each abstract: each paper’s abstract notes the other three as companion papers. Reads as “this is one of four independent angles on the same question.” Captures most of the Four Pillars benefit without the all-or-nothing risk of Option B.

Keeper recommendation: Option C. Each paper Cal-gate-passes on its own merits; the Four Pillars framing in each abstract creates external coherence without forcing simultaneous release. Cal’s external-survivability discipline applies per-paper; the Four Pillars reads as architecturally coherent at the abstract level.

Standing-program connection to substrate-as-thinking

The Four Pillars framework is structurally consistent with this morning’s substrate-as-thinking refinement (T2385):

- Arithmetic channel (Paper #109) = the substrate’s counting structure (how many BST integers exist, where they come from combinatorially)
- Geometry channel (Paper #121) = the substrate’s positional structure (where the integers anchor across classical-math objects)

- Physics channel (Paper #122) = the substrate’s communication structure (what the integers DO as RS-coded alphabet)

If the active-substrate hypothesis ever gets its falsifier (T2385 Level 2 promotion path), the Four Pillars becomes “four sides of the substrate’s thinking” — counting (arithmetic) + positioning (geometry) + encoding (physics) + parity (topology). The Four Pillars framing aligns with the deeper ontological claim Casey was developing.

Coordination roles

Item	Owner	Status
Paper #109	Lyra	Standing for Lyra
tier-discipline read-pass		capacity
Paper #121 v0.2	Grace + Lyra (K57	In progress
substantive	ratified)	
Paper #122 §3 two-level	Grace	Done Tuesday
framing		
Four Pillars abstract	Lyra (theory) + Grace	Pending
coordination (per Option	(catalog) joint	
C)		
Cal gate-pass on each	Cal	Available at his cadence
External strategy update	Keeper	Filed today
(this doc)		
Substrate-as-thinking	Whoever pulls T2385	Open
cross-link	follow-up	

Falsifier framing for the Four Pillars

Each paper has its own falsifiers (per discipline). The Four Pillars META-FALSIFIER:

If a single classical-math source produces all five BST integers + $N_{\max} = 137$ + the specific BST observables in BST primary form, the Four Pillars’ “four independent channels” claim is reduced. Currently no such single source is known; the Four Pillars convergence is non-trivial.

This is the structural fact worth emphasizing externally: each paper exhibits a DIFFERENT mechanism (counting vs geometric vs information-theoretic) producing the same integers. Reductionist external attacks need to invalidate ALL THREE mechanisms, not just one.

Risk flag — what could weaken the Four Pillars

1. Paper #109 counting argument over-asserts without explicit W3-pass classical-math anchor → Four Pillars “arithmetic pillar” weakens; mitiga-

tion = Lyra read-pass before submission

2. Paper #121 Bridge Object tier walks back under future Cal audit → Four Pillars “geometry pillar” weakens; mitigation = K57 ratification stands, Cal PASS Wednesday, no walk-back foreseen
3. Paper #122 active-substrate Level 2 over-pitches beyond Level 1 D-tier evidence → Four Pillars “physics pillar” weakens; mitigation = two-level disclosure stays load-bearing in §3.0
4. Paper #119 topology-pillar derivation fails Cal audit → Four Pillars “topology pillar” weakens; mitigation = Cal PASS Wednesday, K65 ratifies T2395 unified mechanism via Möbius parity ODD
5. External reader rejects Four Pillars framing as marketing rather than substantive → most serious risk; mitigation = Four Pillars convergence is the substantive claim, not a presentation device; emphasize the meta-Type-C structural evidence shape + structural fit (Four = rank²)

Action items

1. Lyra: Paper #109 tier-discipline read-pass when capacity permits (~1-2h)
2. Grace: Paper #121 v0.2 substantive completion + Paper #122 §3 verification post-K57
3. Lyra+Grace coordination: draft Four Pillars abstract framing (Option C) — one paragraph per paper for cross-reference in each abstract; ~30 min joint work
4. Cal: gate-pass each paper at his cadence; emphasize trio-coherence in the per-paper external-survivability check
5. Keeper (me): maintain this coordination document as trio status evolves; flag any drift between papers

Filing status

This document = the Four Pillars external-strategy coordinator. Companion to Grace’s foundational `BST_Three_Papers_Trio_Architecture_Mapping.md` (Phase 1 historical scoping, retained for provenance). Both reference each other: - Grace’s foundational doc: original three-pillar structural mapping (Phase 1) - This doc: Four Pillars external-presentation strategy with topology pillar added (Phase 2 current) - This doc: external-presentation strategy + coordination + risk flags

Together they form the coordination layer for the trio’s external presentation.

— Keeper, 2026-05-19 Tuesday 12:35 EDT